

Table 2. Differentiating features between Neuroretinitis and other closely resembling entities.

|                         | Neuroretinitis                                          | Papillitis                                                                     | Papilledema                                                                                   | CRVO                                                               | AION                                               |
|-------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------|
| VA                      | 6/60-6/12                                               | Light perception to 6/12                                                       | No visual loss                                                                                | Moderate to severe visual loss                                     | Moderate to severe impairment                      |
| Pupillary reactions     | Relative afferent pupillary defect (RAPD+)              | RAPD+                                                                          | Normal                                                                                        | Normal/RAPD+                                                       | RAPD+                                              |
| Laterality              | Unilateral, rarely bilateral                            | Unilateral/bilateral                                                           | Always bilateral                                                                              | Unilateral                                                         | Typically unilateral                               |
| Eye pain                | Nil                                                     | Pain especially on upgaze                                                      | Nil                                                                                           | painless                                                           | painless                                           |
| Visual fields           | Centrocaecal scotoma                                    | Central/centrocaecal scotoma                                                   | Enlarged blind spot                                                                           | Normal                                                             | Altitudinal field defects                          |
| Color vision            | Severely impaired, disproportionate to visual loss      | Severely impaired, disproportionate to visual loss                             | Normal                                                                                        | Normal                                                             | Diminished in proportion to level of v/a           |
| Systemic symptoms       | Fever, rash                                             | Weakness of limbs                                                              | Headache, vomiting                                                                            | -                                                                  | Headache, jaw claudication, polymyalgia rheumatica |
| Fundus findings         | Disc swelling of 2D, hyperemic with macular star figure | Disc swelling rarely above 2D, venous engorgement and haemorrhages less marked | Disc swelling frequently higher, upto 8-9D, more venous engorgement, macular star may develop | Disc edema, macular star along with haemorrhages and soft exudates | Pale disc edema                                    |
| Fluorescein Angiography | Leakage from disc and peripapillary retina              | Leakage from disc and peripapillary retina                                     | Leakage from disc and peripapillary retina                                                    | Shows areas of capillary non-perfusion                             | Unequal choroidal filling in arterial phase        |
| VEP                     | Decrease in amplitude, increase in latency              | Decrease in amplitude, increase in latency                                     | Normal                                                                                        | -                                                                  | Decrease in amplitude                              |
| Specific associations   | Syphilis, cat-scratch disease, Lyme's disease           | Multiple sclerosis                                                             | -                                                                                             | hypertension, diabetes, glaucoma                                   | Giant cell arteritis, hypertension, diabetes       |
| Prognosis               | Good                                                    | Good                                                                           | Good with relief of raised ICT                                                                | Depends on initial visual acuity                                   | Poor                                               |

disc swelling that may on occasion be associated with the development of a macular star figure. These mimicking conditions include papilledema, anterior ischemic optic neuropathy, and infiltration of the optic disc by tumor.<sup>14</sup> Systemic hypertension may also produce both optic disc swelling and a macular star figure. The disk edema and retinopathy resolves after the hypertension is controlled.<sup>29</sup> Optic disc swelling in patients with systemic vascular disease like diabetes and hypertension can be differentiated from neuroretinitis by the absence of abrupt visual loss, background retinopathy and a medical history of such

conditions. Spontaneous resolution of the disc edema and recovery of visual acuity serve as distinguishing features of neuroretinitis from papilledema and ischemic optic neuropathy.

### Investigations: (Table 3)

Investigation into the etiology of neuroretinitis should begin with a careful history including questioning regarding sexually transmitted diseases, cat-scratches, skin rashes, tick bites, lymphadenopathy, fever, and flu-like illnesses. Complete physical and ocular